

EE514 Final Report

Introduction:

This project consisted of a 2.4 - 2.6 GHz VNA design. All design files for the project are in this public repository [fdreshaj/VNA_Design](#) (Needs to be updated to the latest RF coupler designs)

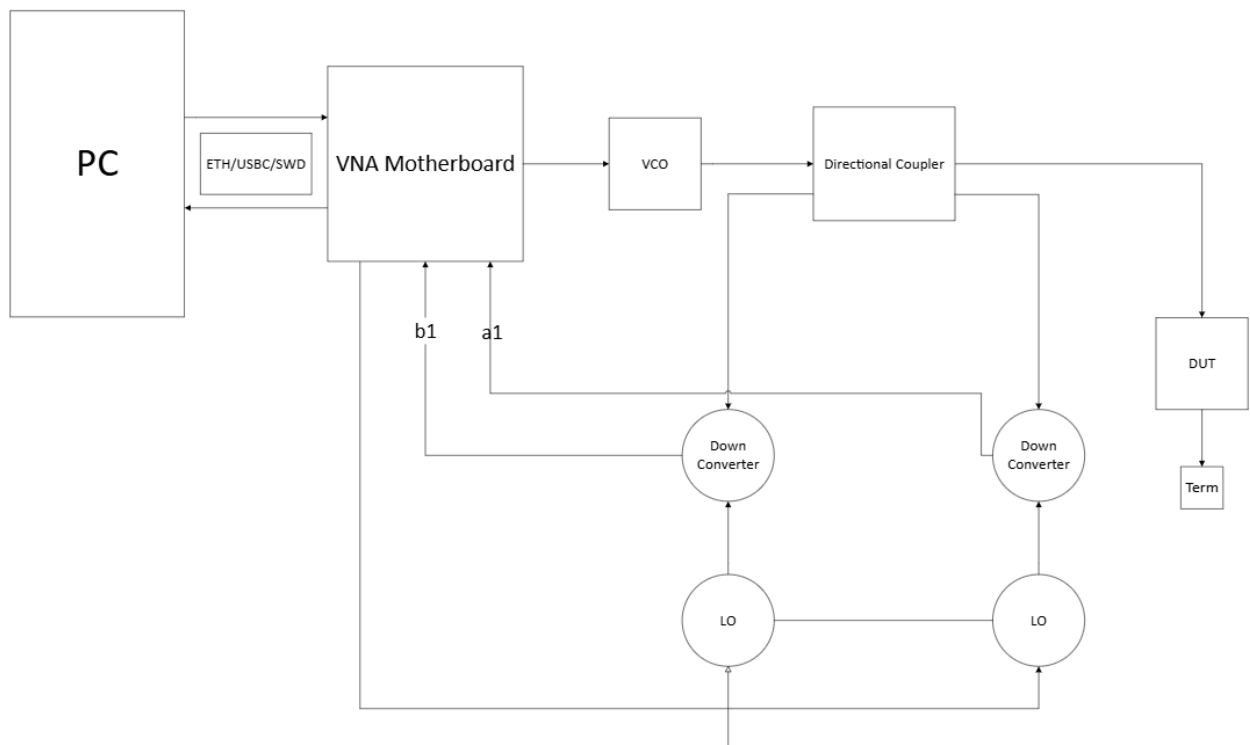
Deliverables:

2.4 - 2.6 GHz VNA. This will include RF designs of a 2.4 - 2.6 GHz VCO, Directional coupler (Will redo in CST), mixer down converter (Will redo in CST) and non RF subsystems.

Theory of Operations:

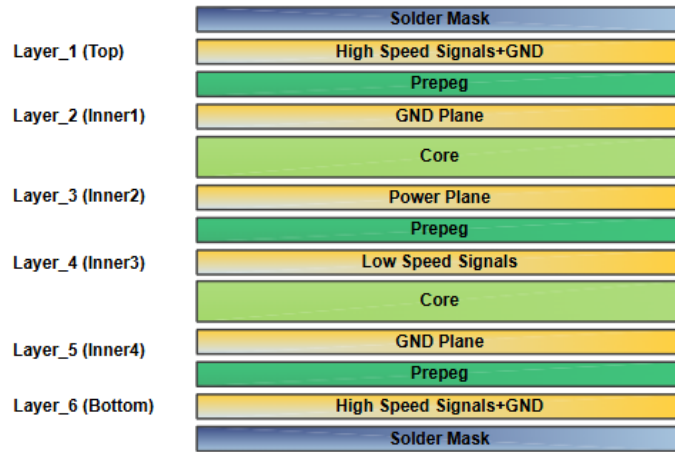
To gather S-parameter data, more specifically a, b wave data the VNA motherboard will generate a 2.3-2.8 GHz sweep through to the directional coupler, which will send a through wave to the DUT and a downconverter. The reflection from the DUT will go back into the coupler sending a through wave to the other downconverter. The outputs of the downconverter will be sampled through using a 20MSPS ADC which will be stored in flash memory or RAM then sent back to the PC via ETH.

Subsystems Overview:



VNA Motherboard:

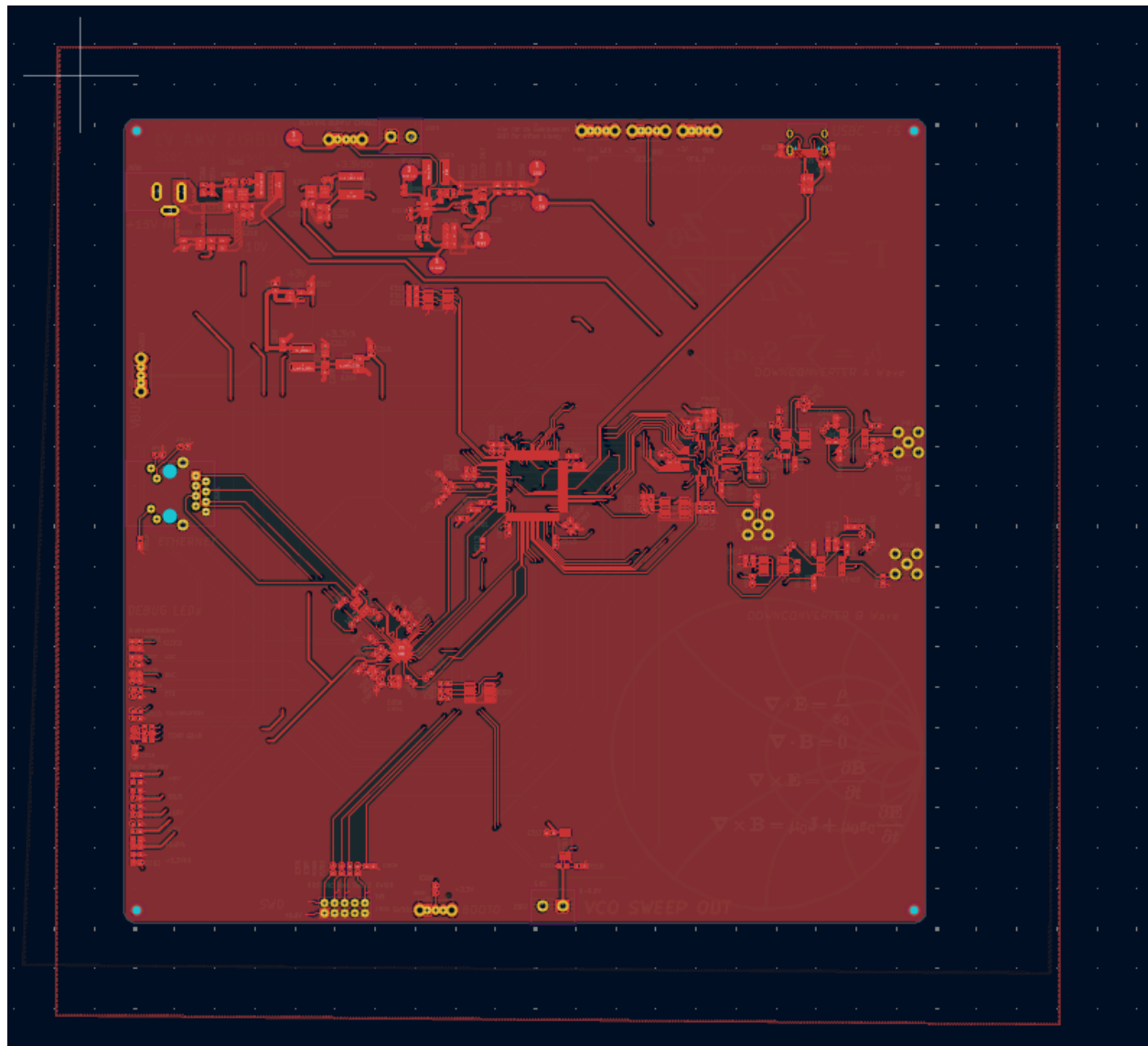
Stackup

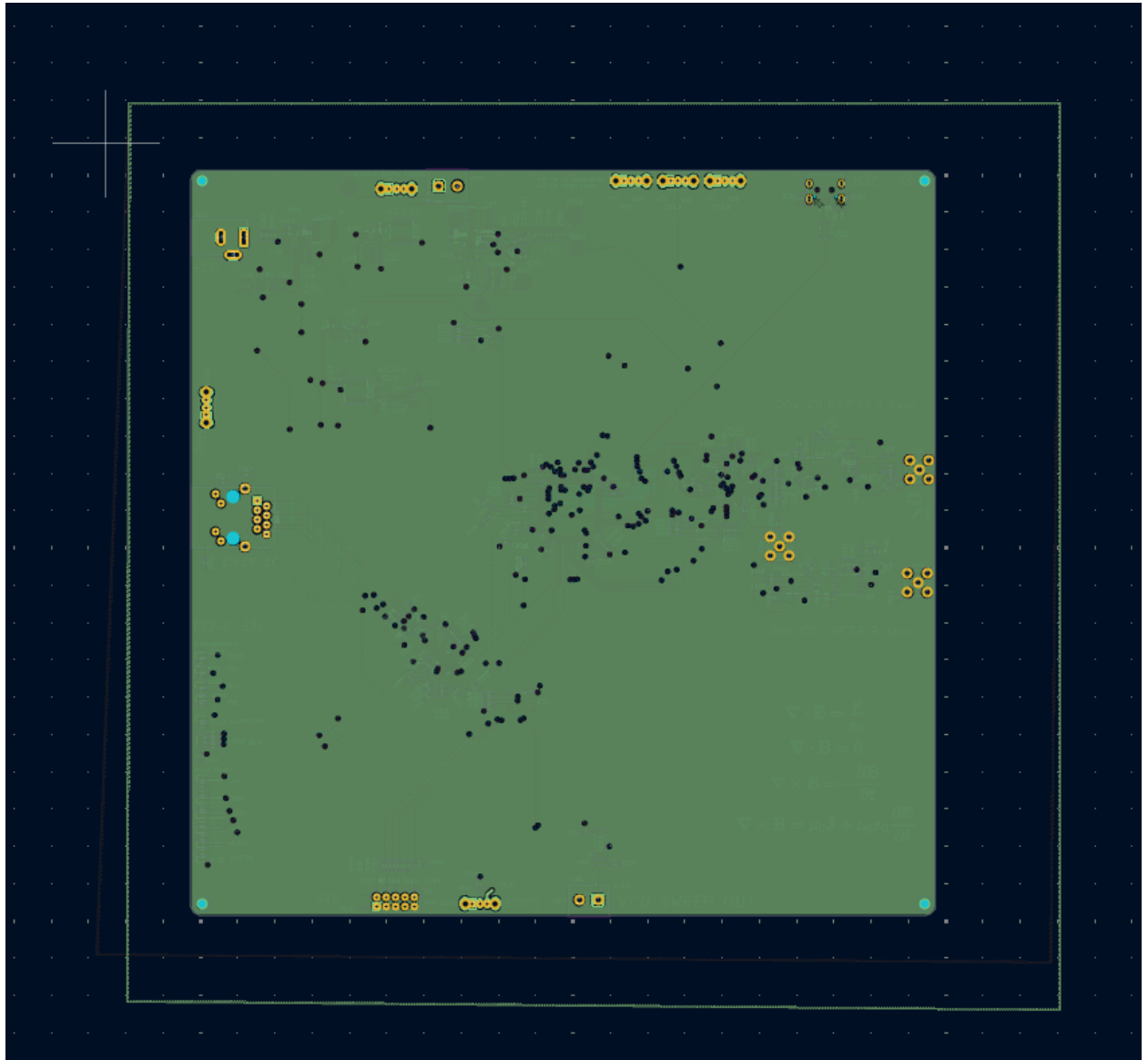


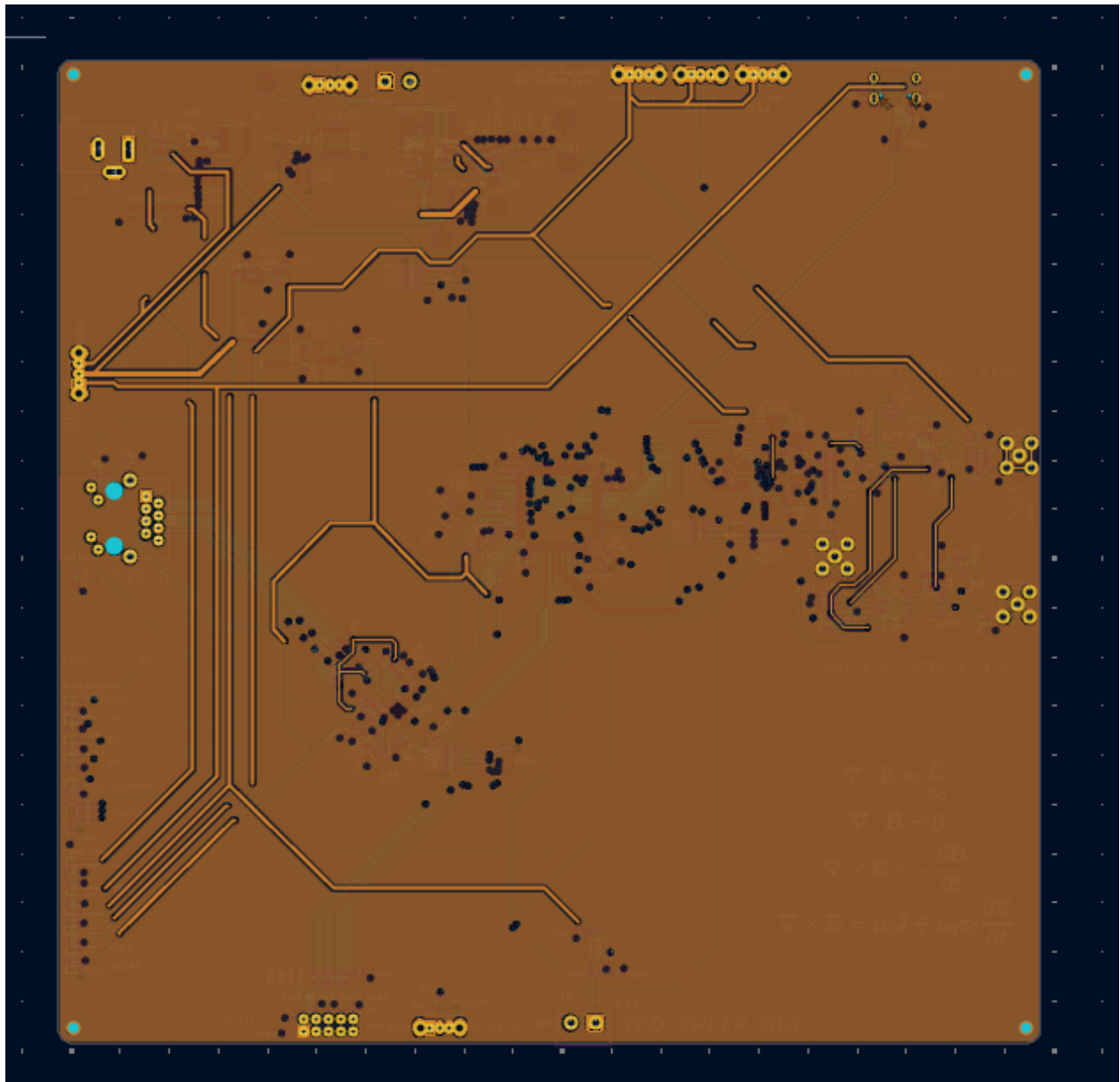
(STM32H7 Datasheet)

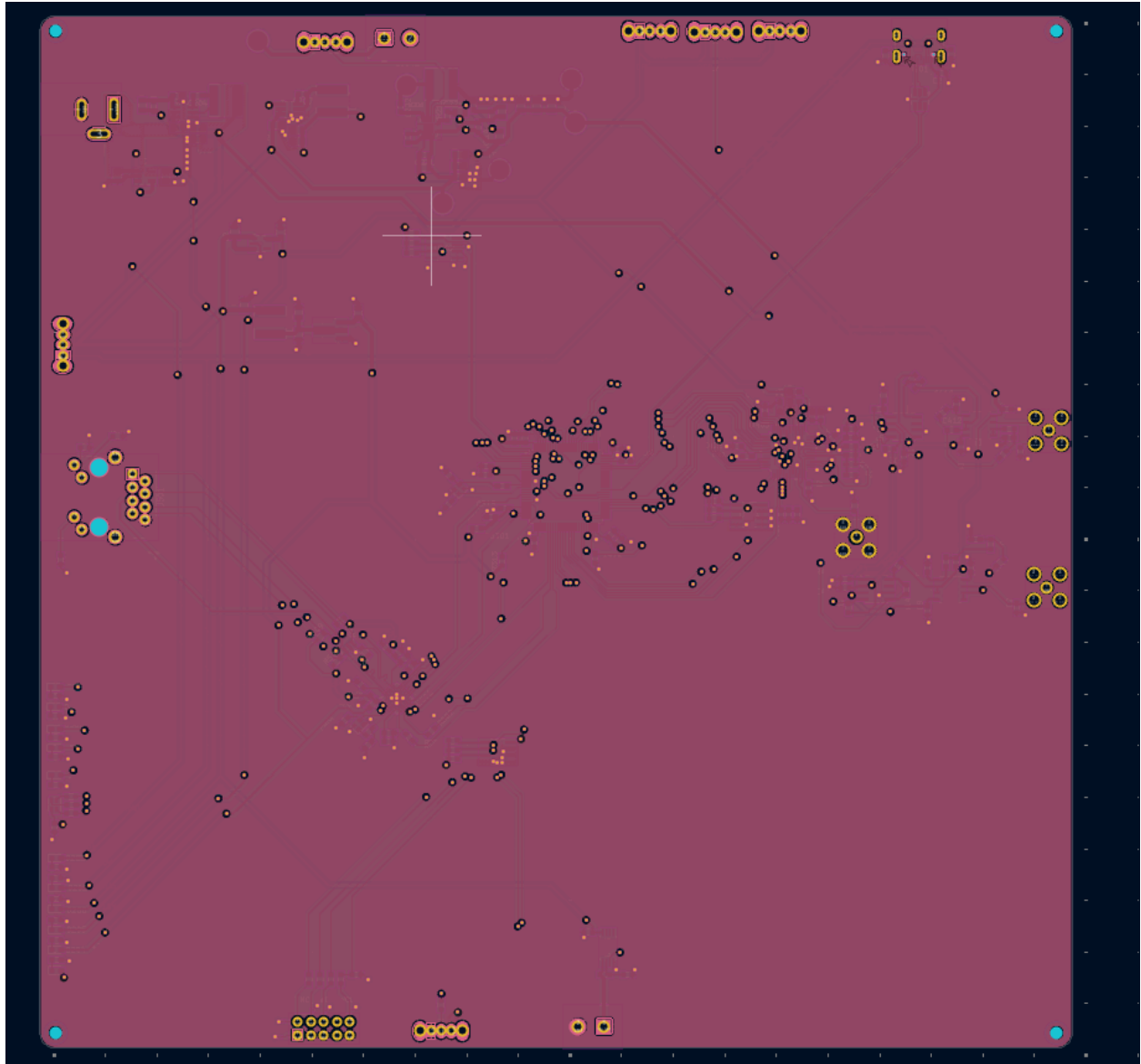
I went with this stackup since it is recommended on the datasheet, it also gives me enough spacing to layout all of my ADC pins into the microcontroller which would be more difficult to accomplish on a 4 layer board.

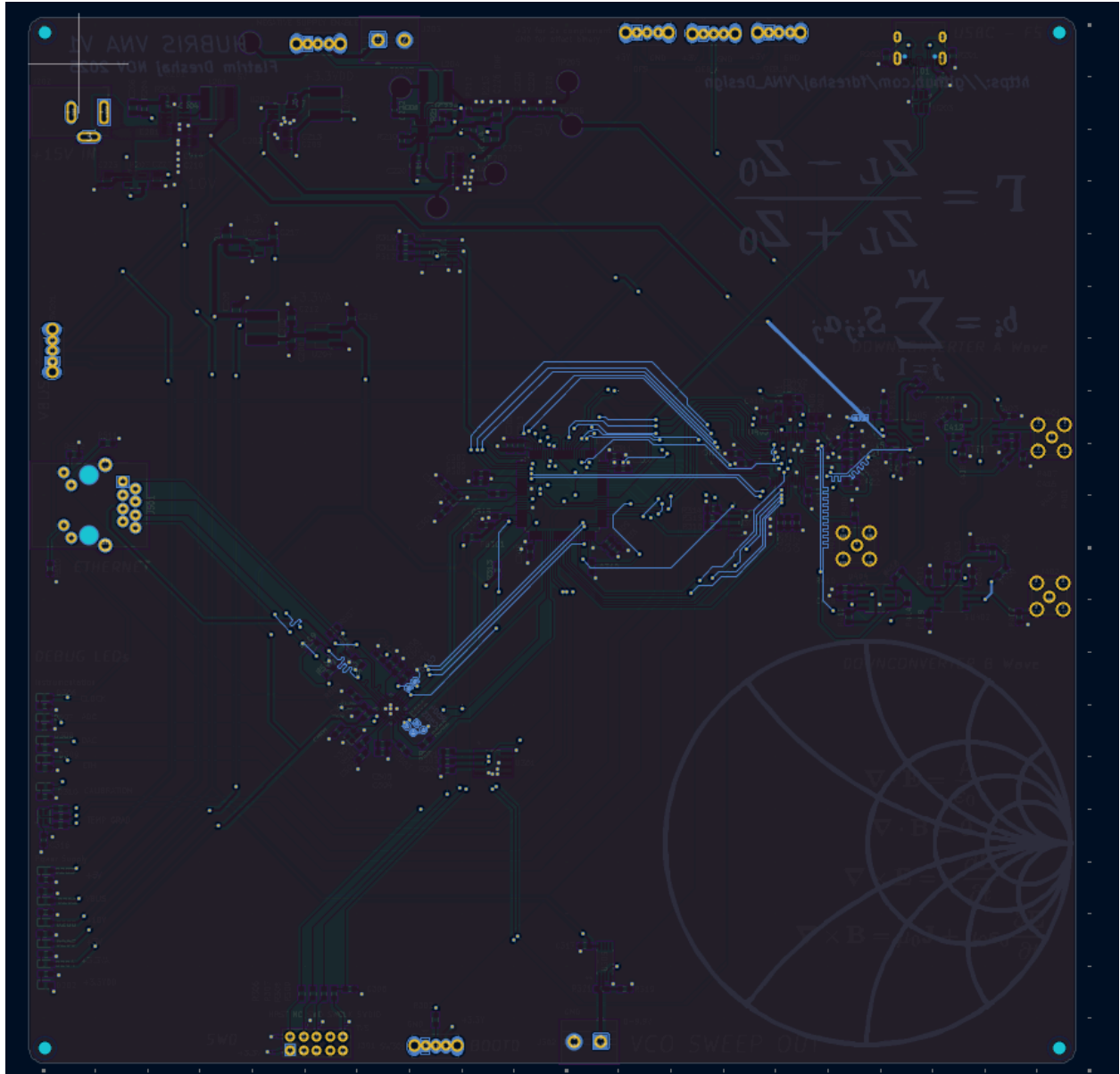
Layer View



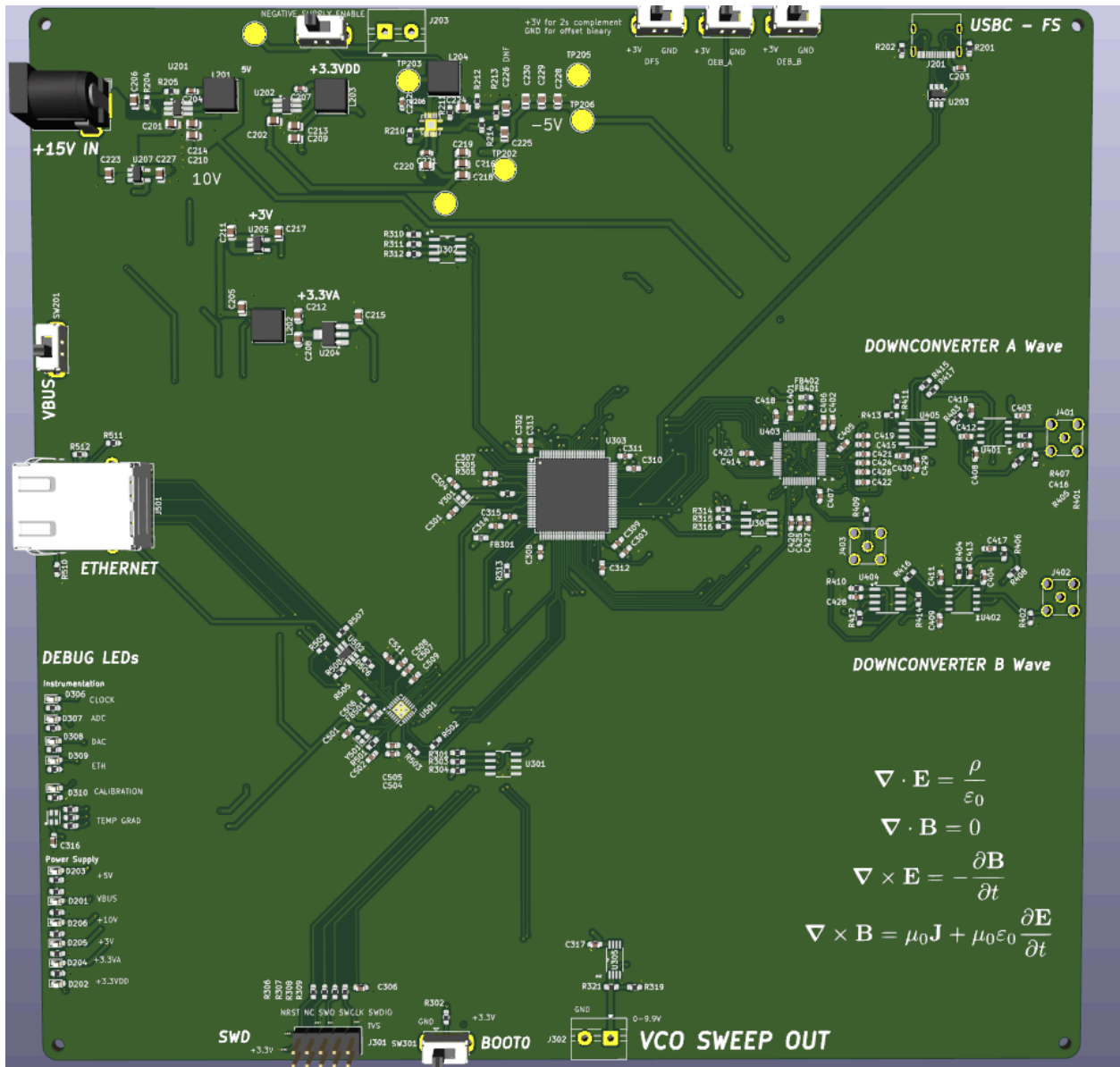






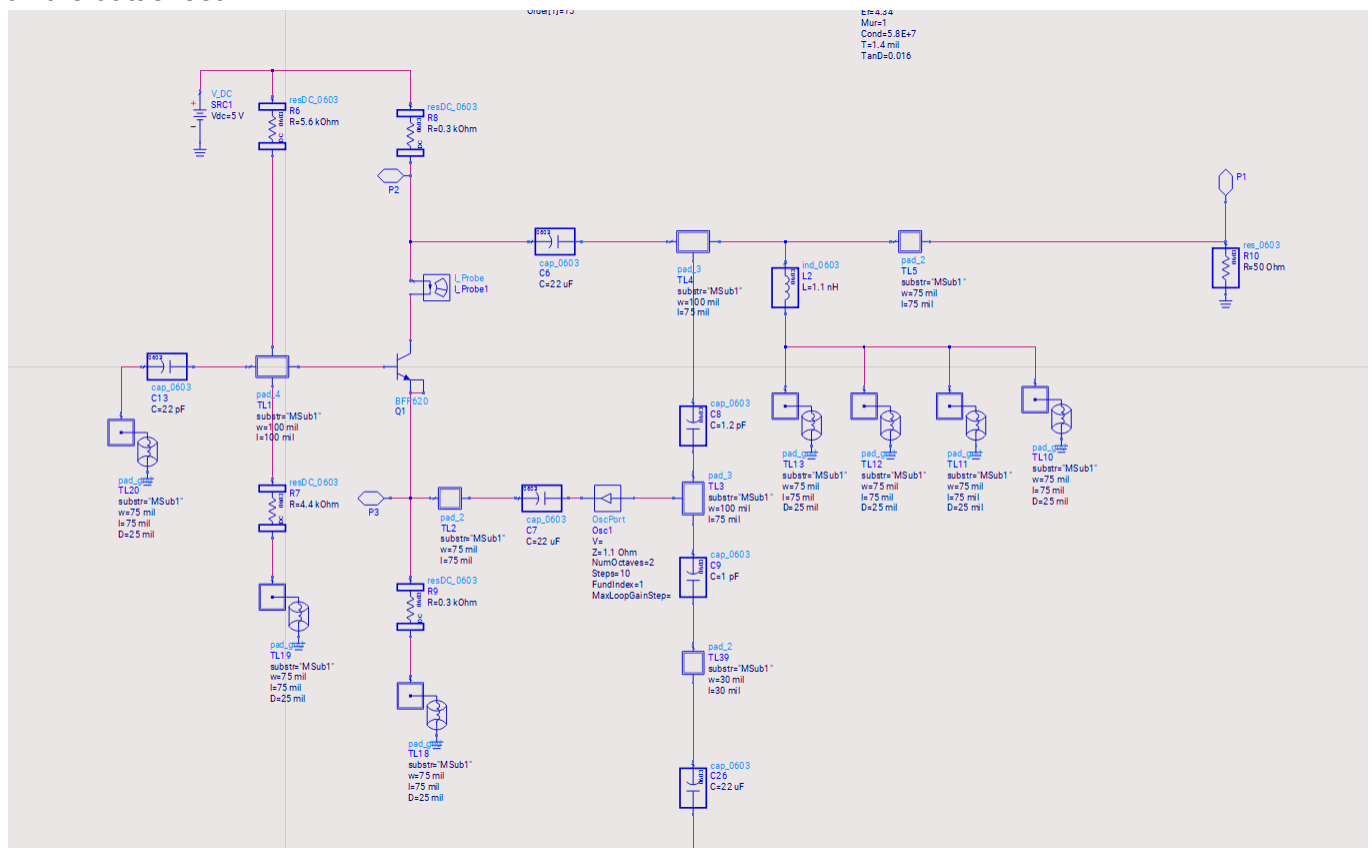


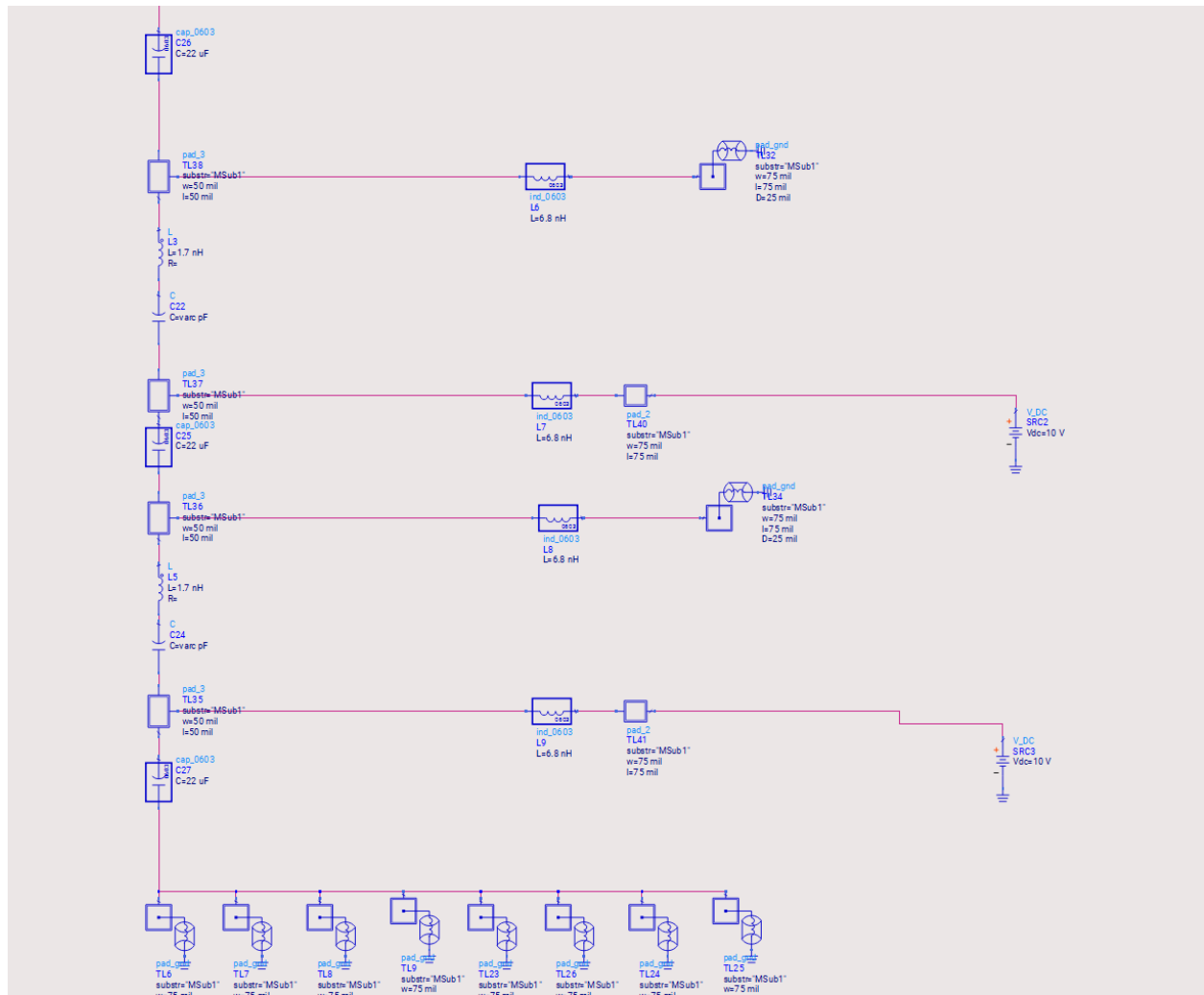
3D View:



VCO:

For the VCO I'm using a Modified Colpitts Common Base Design. I am biasing at 5mA based on the minimum noise figure of the transistor. I'm controlling the frequency through the use of two SMV 1247 varactor diodes that provide me with a 0.7pF capacitance base and a capacitance ratio of 10 meaning I can reach up to 7 pF capacitance on each. I chose this diode specifically because I swept the modified colpitts architecture I had with two capacitors in series and found that from values of 0.7-7 pF I can get through my frequency design specifications. Additionally I did not have the diode spice model so I simply used the equivalent model stated on the datasheet





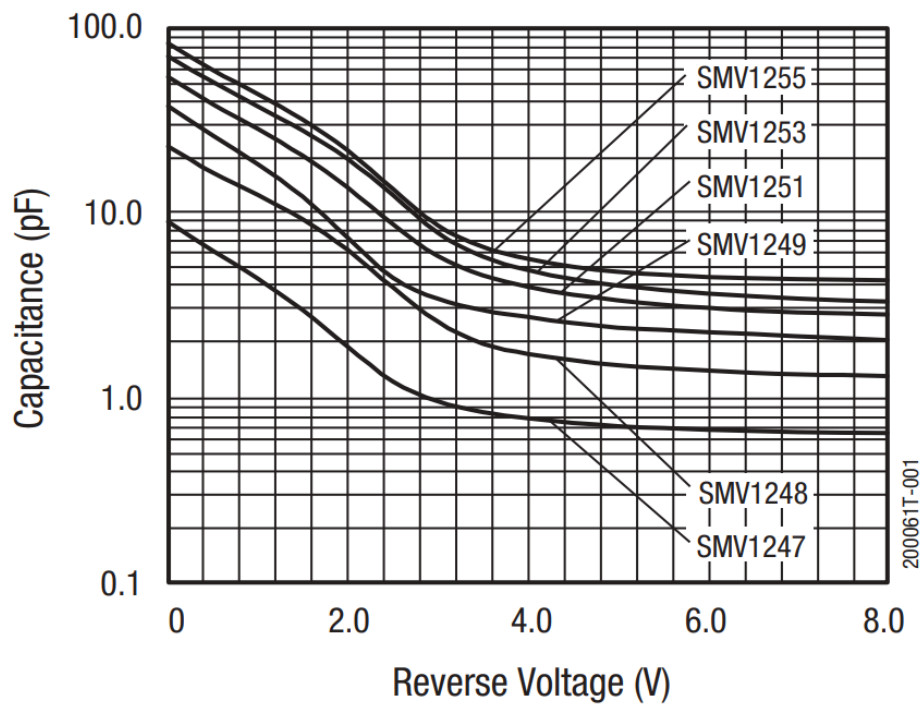
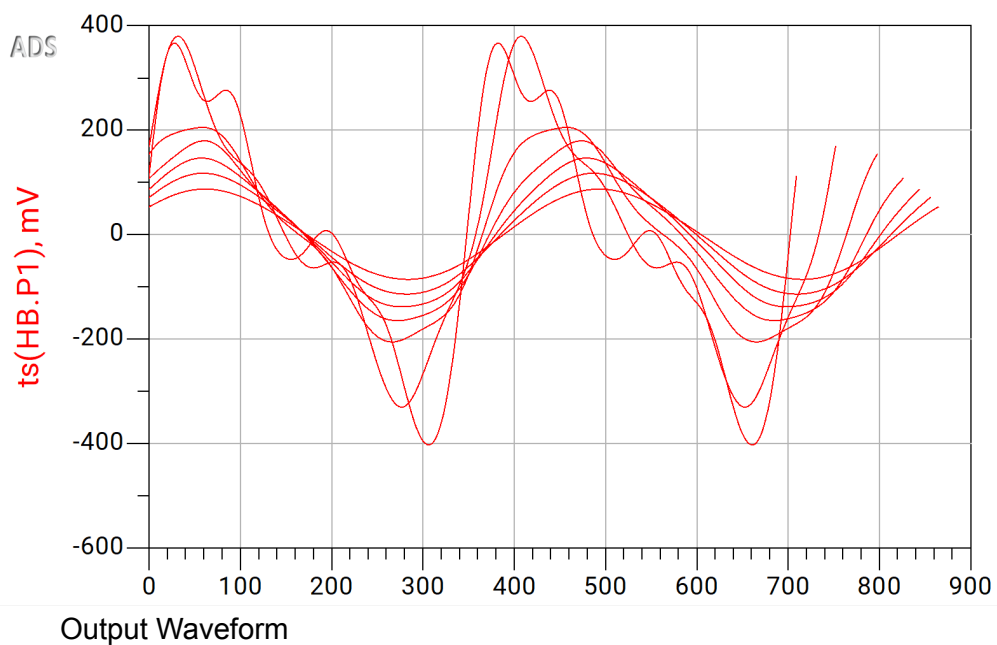
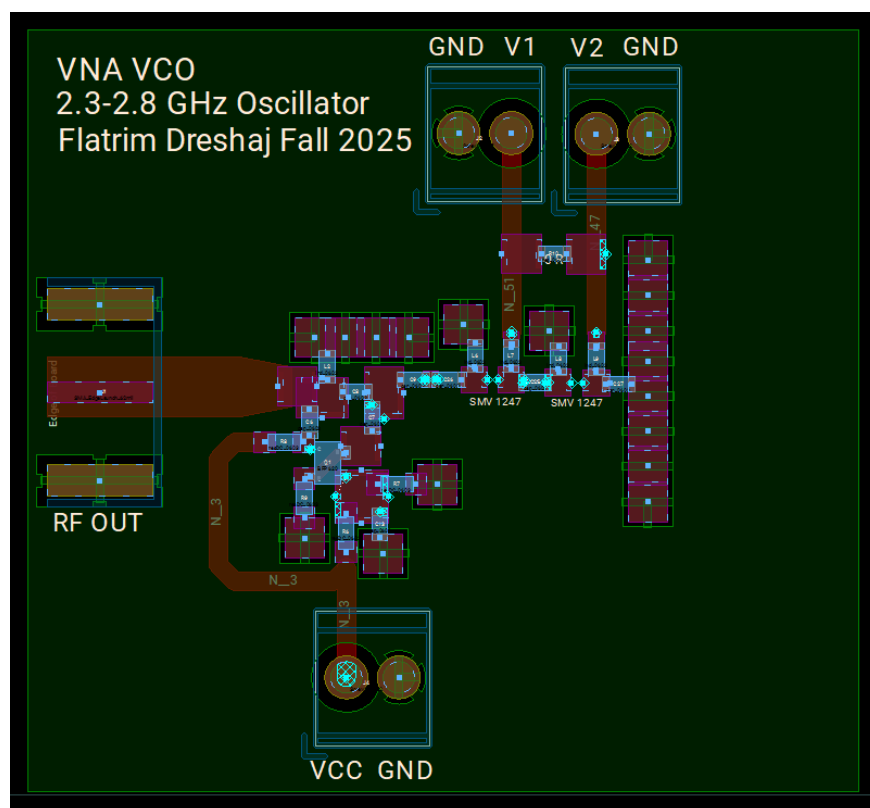
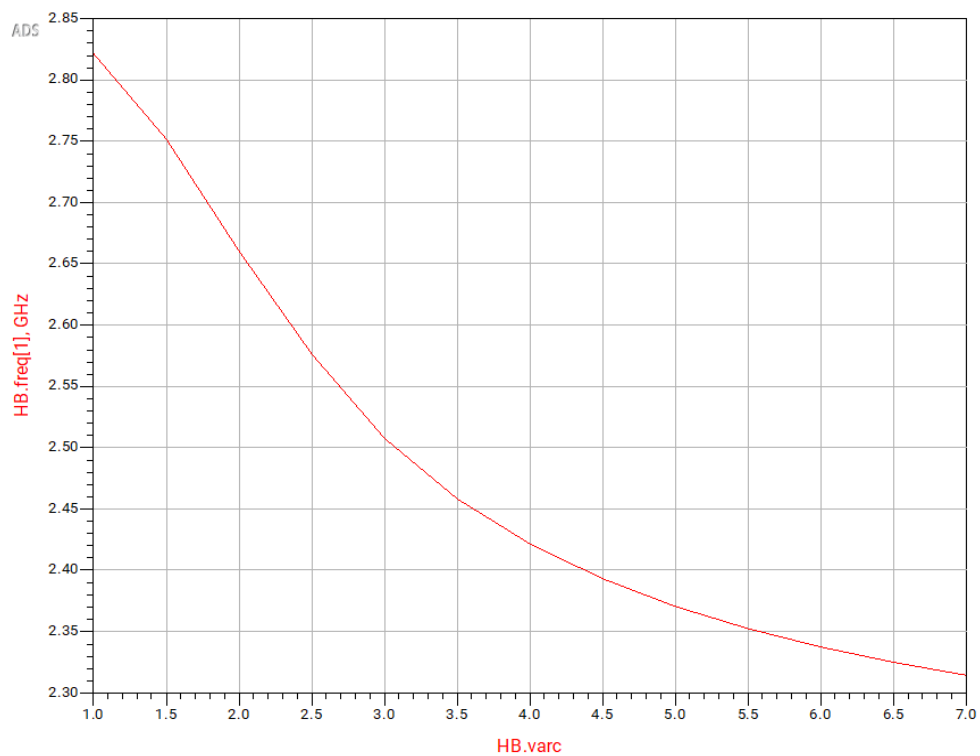


Figure 1. Capacitance vs Reverse Voltage



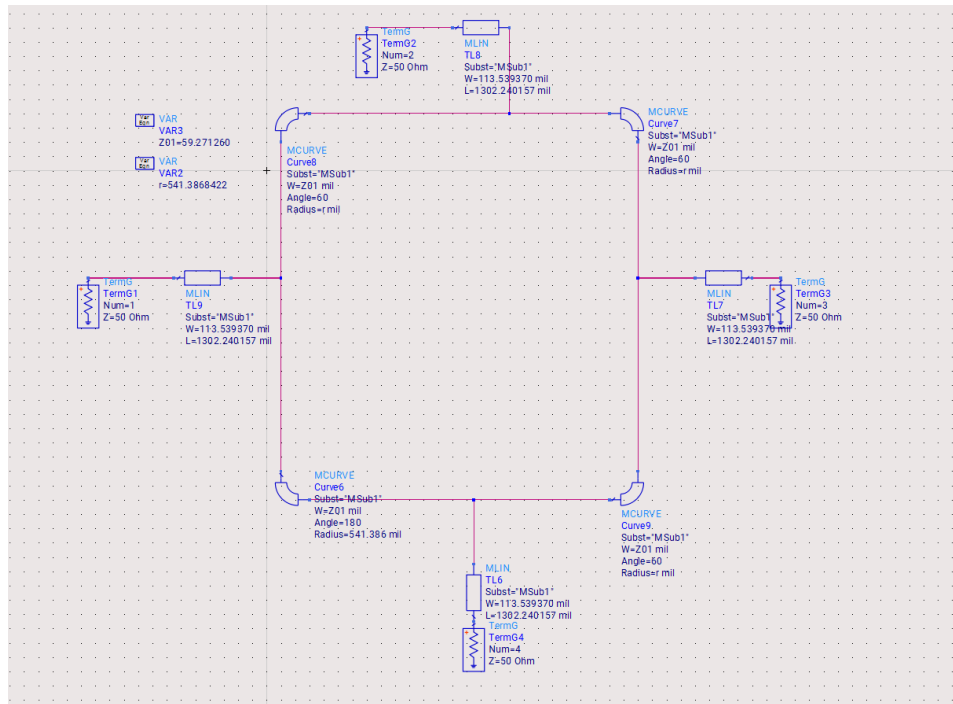


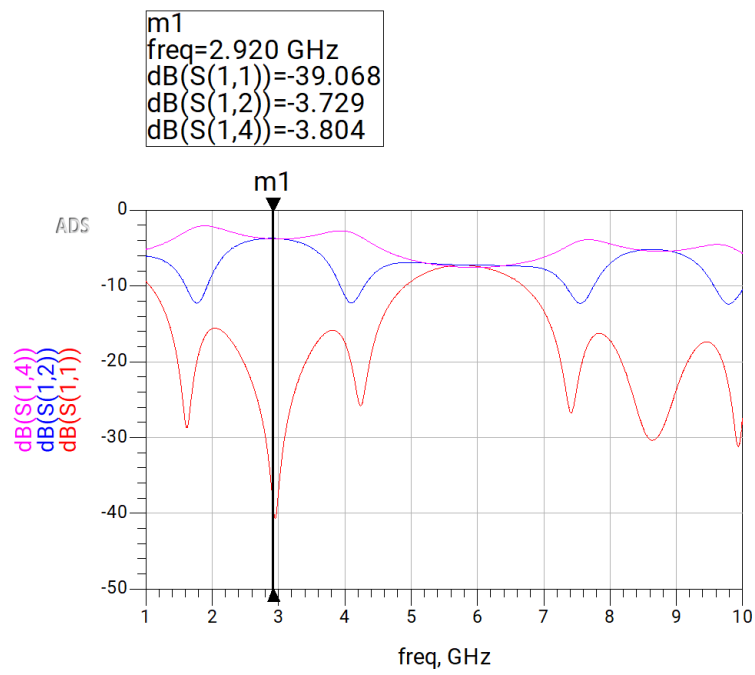
Directional Coupler design:

For the directional coupler I had to choose between 90 degree couplers and 180 degree couplers. To implement these two couplers I designed multiple branch line couplers, rat race couplers and modified rat race couplers to see what gives me the best performance. Ultimately I ended up using a rat race coupler and a modified rat race coupler.

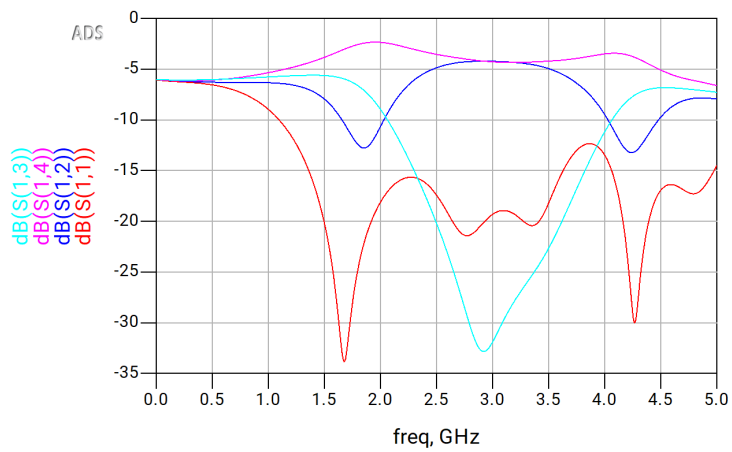
For my design equations for the rat race I used a $Z_{design} = 70.74 \text{ Ohms}$ and since I am using curved lines I needed to find the radius required to give the $\lambda/4$ and $3\lambda/4$ circumference. To find that I simply used the arc length formula and solved for r, $\frac{\theta}{360} 2\pi r = \lambda/4$ and $\frac{\theta}{360} 2\pi r = 3\lambda/4$.

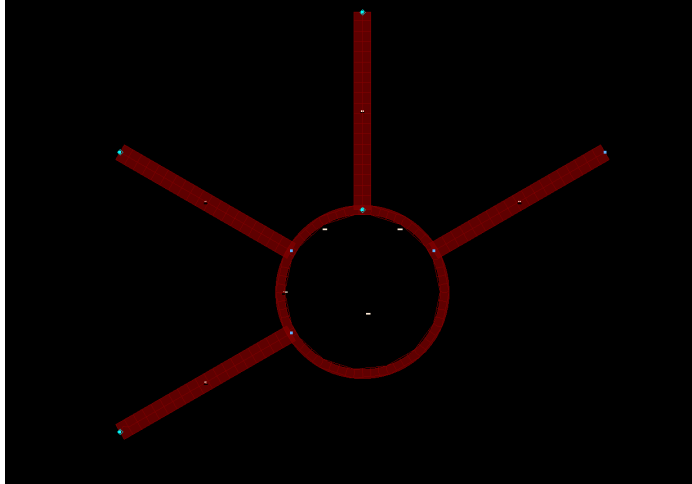
For the modified rat race, instead of the $3\lambda/4$ curve length we could use a $\lambda/4$ curved parallel coupled line. Finding the even/odd mode impedances can be simply done via multiplying and dividing by 2.414 for Z_{oe} and Z_{oo} respectively.



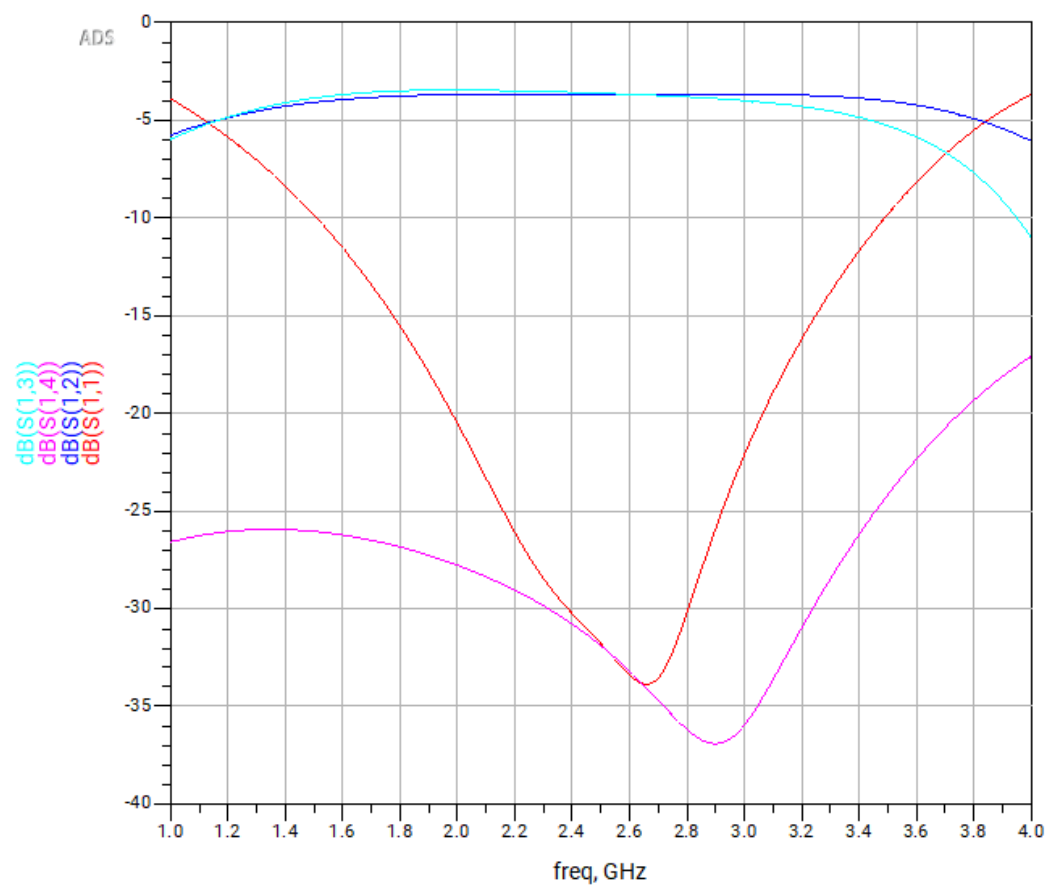


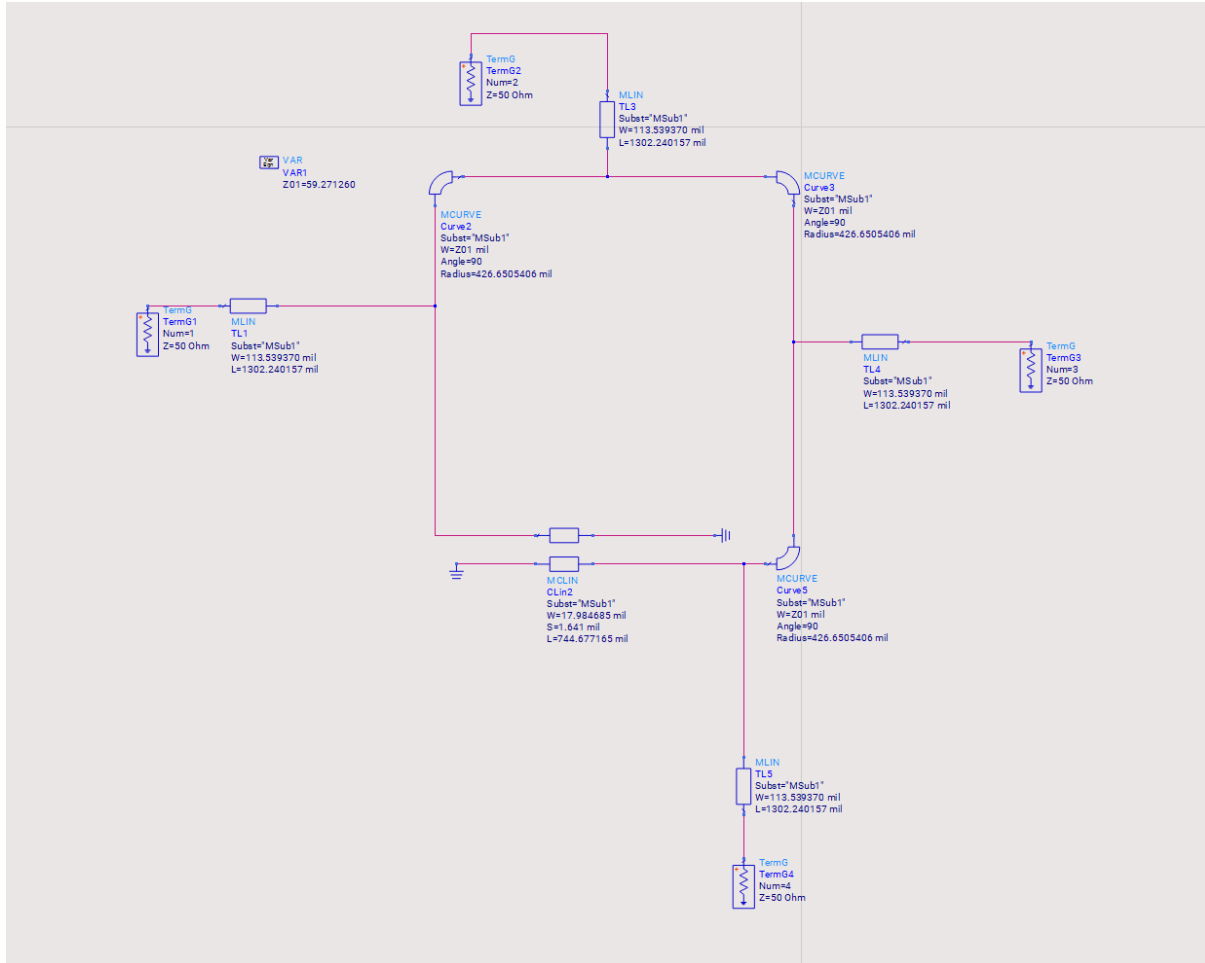
Layout:



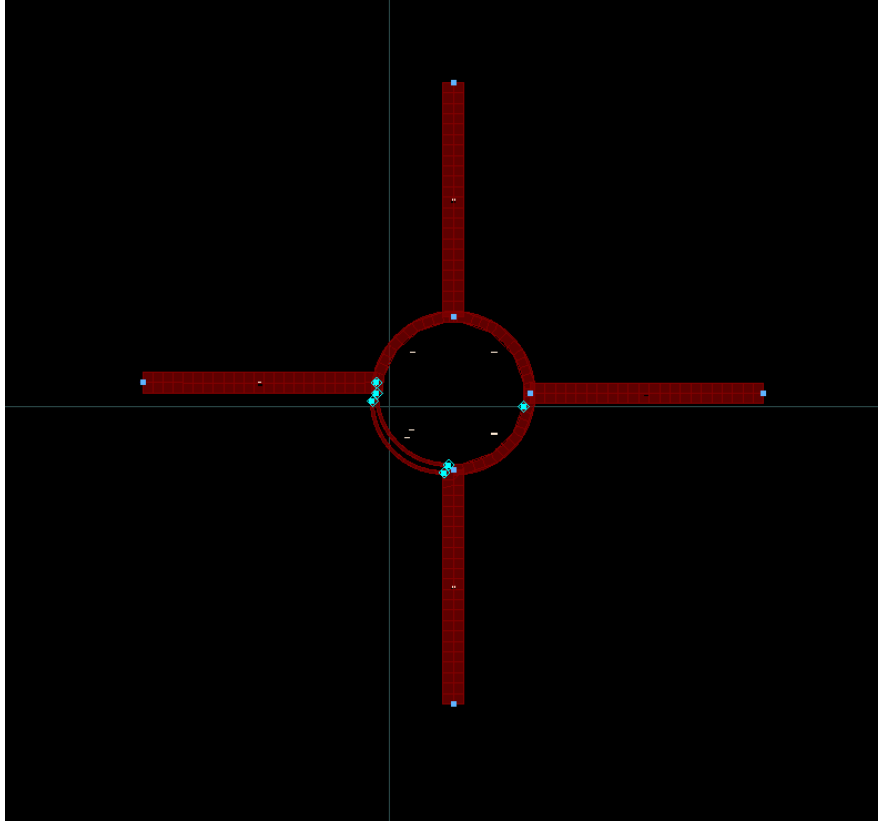


Wideband Coupler Design:





Layout:

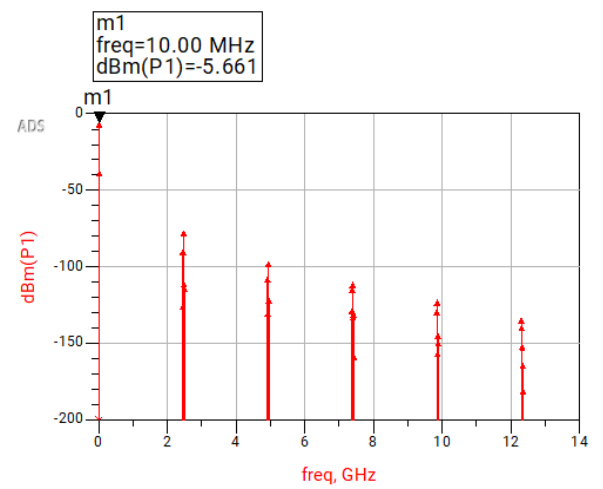
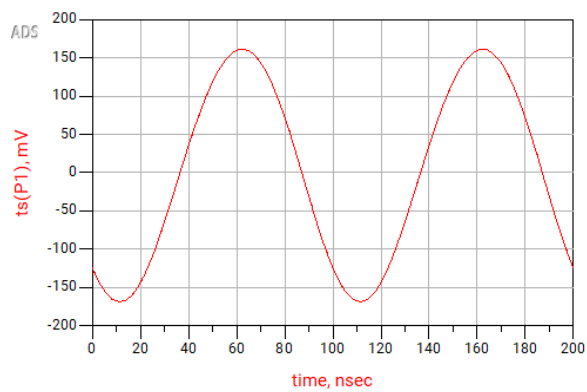
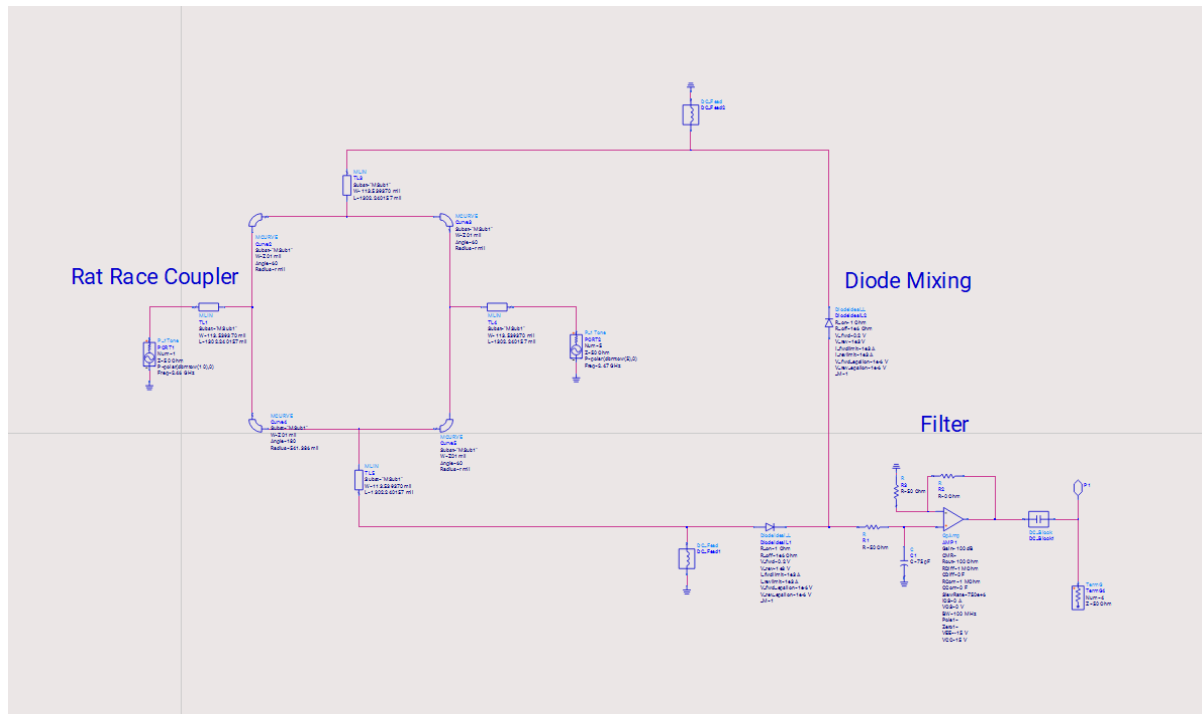


Not really possible to be done in ADS so will continue this design in CST.

Downconverter

For the downconverter design, I went with the simplest in slot solution which was to use a balanced 90 degree or 180 degree mixer. From testing I found that the 180 degree mixer provides better performance for the $\cos(\text{RF-LO})$ harmonics. Giving me close to 6dB of loss on the output. The output of the downconverter I am just filtering out to get the harmonic that I want. The diode choice for this was not as important besides for choosing the desired forward voltage and from testing I found that that was 600 mV. I also added parasitic capacitance and resistance testing on the diodes (Not shown in the photo, in the ADS files).

The output of the mixer will be fed into the anti aliasing filter which then will go into the differential transformer for the ADC turning the input IF into a differential signal which will make ADC conversion more accurate (according to the manufacturer) which will then finally be sampled by the dual ADC AD9238 at 20MSPS.



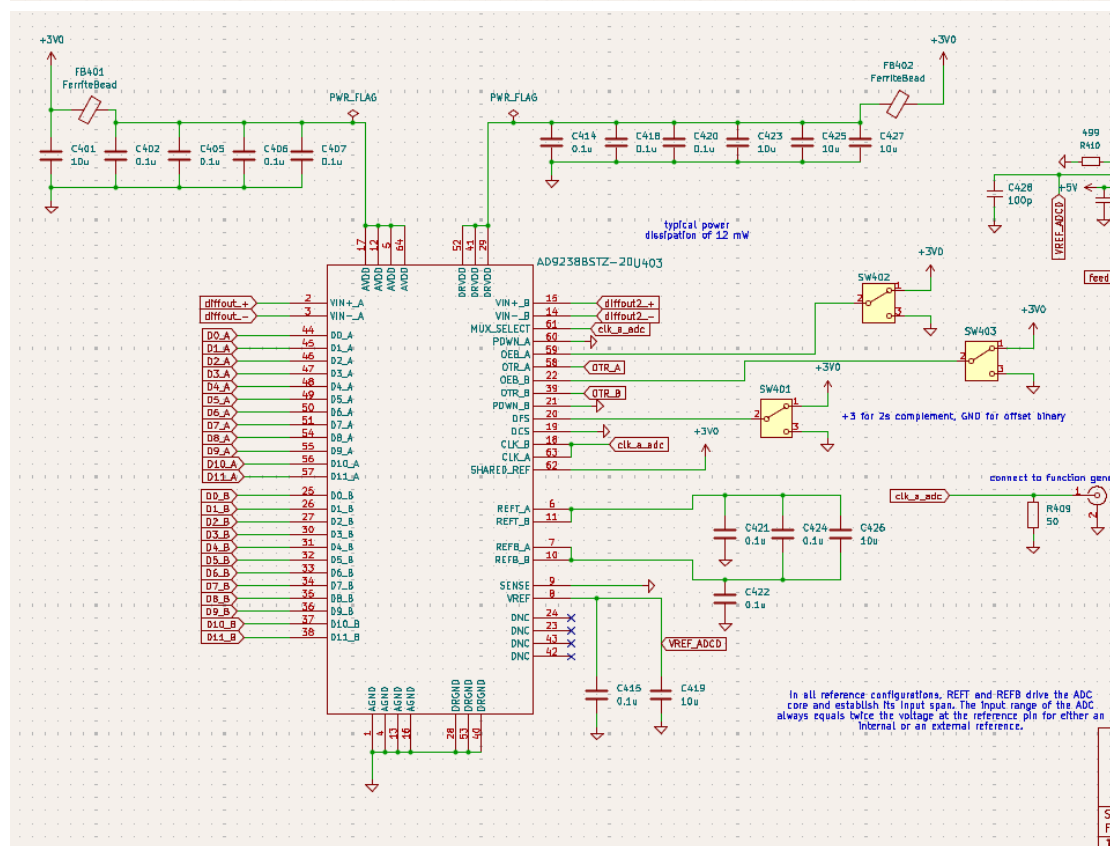
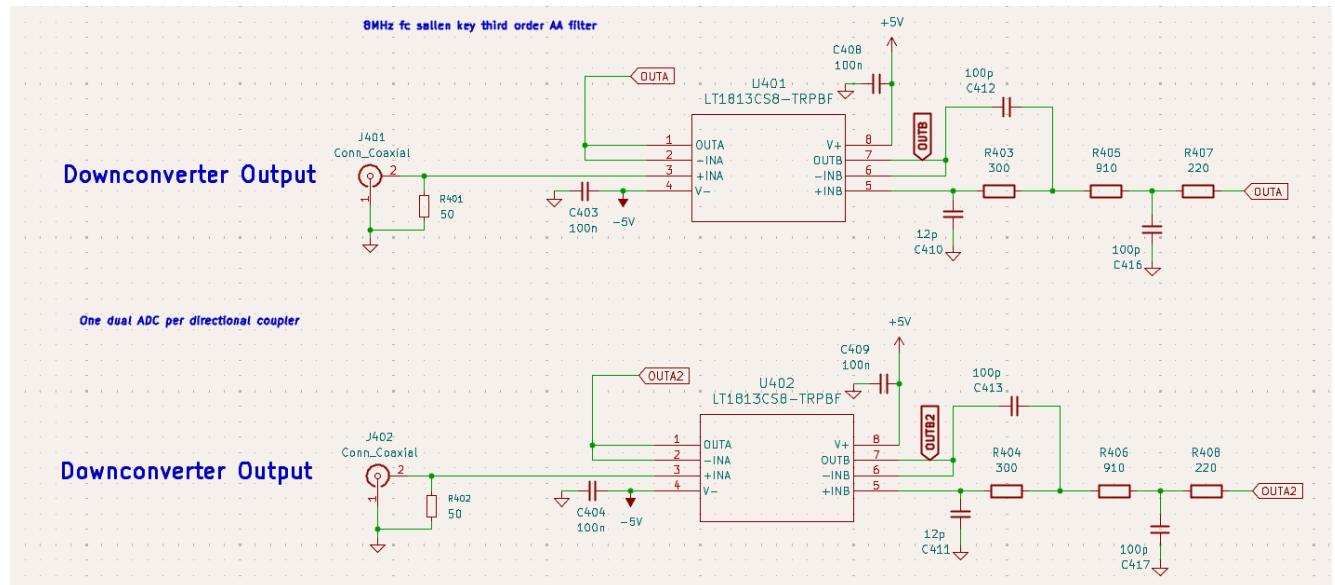


Table 7. Reference Configuration Summary

Selected Mode	SENSE Voltage	Resulting VREF (V)	Resulting Differential Span (V p-p)
External Reference	AVDD	N/A	2 × External Reference
Internal Fixed Reference	VREF	0.5	1.0
Programmable Reference	0.2 V to VREF	$0.5 \times (1 + R2/R1)$	2 × VREF (See Figure 36)
Internal Fixed Reference	AGND to 0.2 V	1.0	2.0

Signal Processing

After transferring the data from the Motherboard to the PC via Ethernet we can simply take the FFT of the signal to get the magnitude and phase information we need to determine S Parameters.

Conclusion

This project was one of the best learning experiences I have had so far in my studies. I got to know much more about RF design, transmission line design, PCB design than any other project I was a part of. Now that most of the design process is done, we will begin to fabricate and test these designs to compare to simulation.